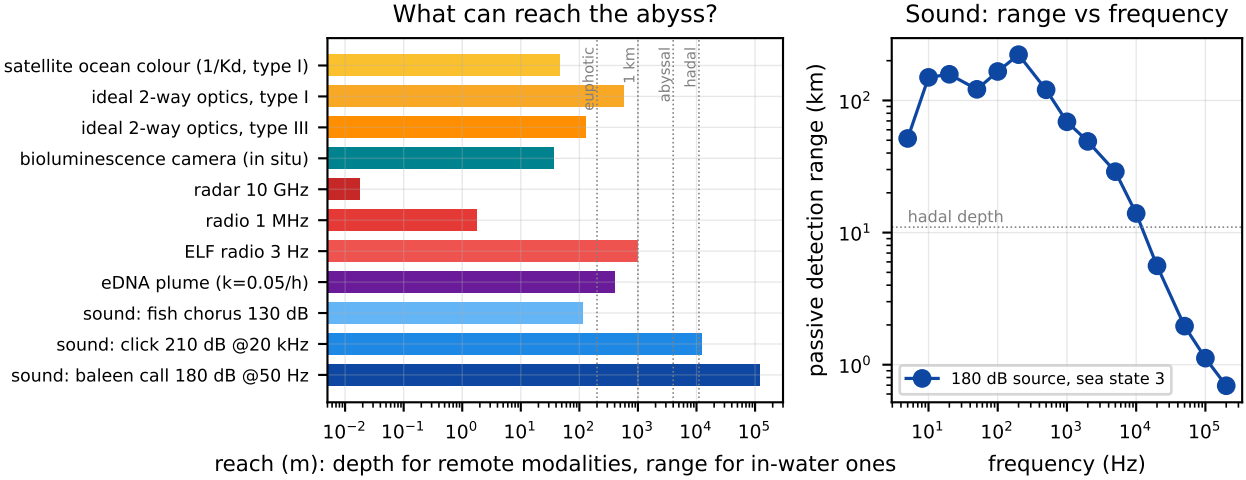
What can reach the abyss?
satellite ocean colour (1/Kd, type I)
ideal 2-way optics, type I
ideal 2-way optics, type III
bioluminescence camera (in situ)
radar 10 GHz
radio 1 MHz
ELF radio 3 Hz
eDNA plume (k=0.05/h)
sound: fish chorus 130 dB
sound: click 210 dB @20 kHz
sound: baleen call 180 dB @50 Hz
euphotic
1 km
abyssal
hadal
10^{-2} 10^{-1} 10^{0} 10^{1} 10^{2} 10^{3} 10^{4} 10^{5}
reach (m): depth for remote modalities, range for in-water ones
Sound: range vs frequency
passive detection range (km)
10^{2}
10^{1}
10^{0}
hadal depth
180 dB source, sea state 3
10^{1} 10^{2} 10^{3} 10^{4} 10^{5}
frequency (Hz)